Fundamentals in Ai & ML

Title : Ai Stress Testing agent(debatrr)

- Contents:

Name - Kartikey Srivastava
Roll No - 25BAI11235
Slot - B11+B12+B13
Faculty - Dr. Vaishnavi S

PROJECT REPORT

- ## <u>Introduction</u> :

debatrr is a simple CLI-based multi-agent system made to test startup ideas. Usually when we think of an idea, we either get too confident about it or reject it too early. This project tries to make that process more structured.

The system basically creates a debate around the idea. One agent supports it, another goes against it, and then a final agent gives insights. The goal is not to give a final answer but to show both sides clearly so the user can think better.

- ## <u>Problem Statement</u> :

Validating ideas properly is actually harder than it looks. Most people either rely on gut feeling or do random research without structure.

Some common problems:

- Bias towards your own idea
- Missing obvious flaws
- No structured comparison of pros and cons
- Time-consuming validation

- ❖ Overthinking or underthinking

So the problem is: *how can we quickly test an idea in a more balanced and structured way?*

And this project is aimed to answer the above question.

- **Objectives :**
  - ❖ Take raw idea input and structure it
  - ❖ Simulate a debate between positive and negative sides
  - ❖ Decide automatically how long the debate should go
  - ❖ Extract useful insights from the discussion
  - ❖ Help user understand strengths and weaknesses of the idea

---

- **Functional Requirements :**

  - ❖ Accept idea input in free text
  - ❖ Convert idea into structured JSON
  - ❖ Run debate between FOR and AGAINST agents

- ❖ Store conversation in a transcript
- ❖ Stop debate dynamically (between 2 to 6 rounds)
- ❖ Generate final insights
- ❖ Allow user to view arguments and insights separately

- **<u>Non Functional Requirements</u> :**

  - ❖ Simple CLI interface
  - ❖ Fast enough responses using model API
  - ❖ Easy to modify and extend
  - ❖ Works with cloud and local models
  - ❖ Not fully reliable due to no external data source

- **<u>System Architecture</u>** :

The system is based on multiple agents working one after another (not parallel).

- Agents involved:

- ❖ Enrichment Agent → converts idea into JSON

- ❖ FOR Agent → supports the idea
- ❖ AGAINST Agent → criticizes the idea
- ❖ Supervising Agent → checks if debate is useful
- ❖ Insight Agent → gives final insights

- Main data structures:

- ❖ transcript → stores all messages
- ❖ for_agent_mem → stores FOR side arguments
- ❖ against_agent_mem → stores AGAINST side arguments
- ❖ verdict → stores final output

- **<u>Design Decisions and Rationale</u>** :

  - ❖ Used sequential flow because it is easier to control and debug
  - ❖ JSON structure helps keep inputs consistent
  - ❖ Dynamic stopping avoids unnecessary long debates
  - ❖ Memory lists help agents remember previous points
  - ❖ CLI was chosen because building UI was not needed for this stage

---

- ## **Implementation Details** :

- Tech stack used:

  - Python
  - Ollama module
  - gpt-oss:120b-cloud model
  - JSON module

- Setup:

  - Install Ollama
  - Use cloud model or local model (qwen3.5:9b)
  - Install required packages
  - Run the Python file

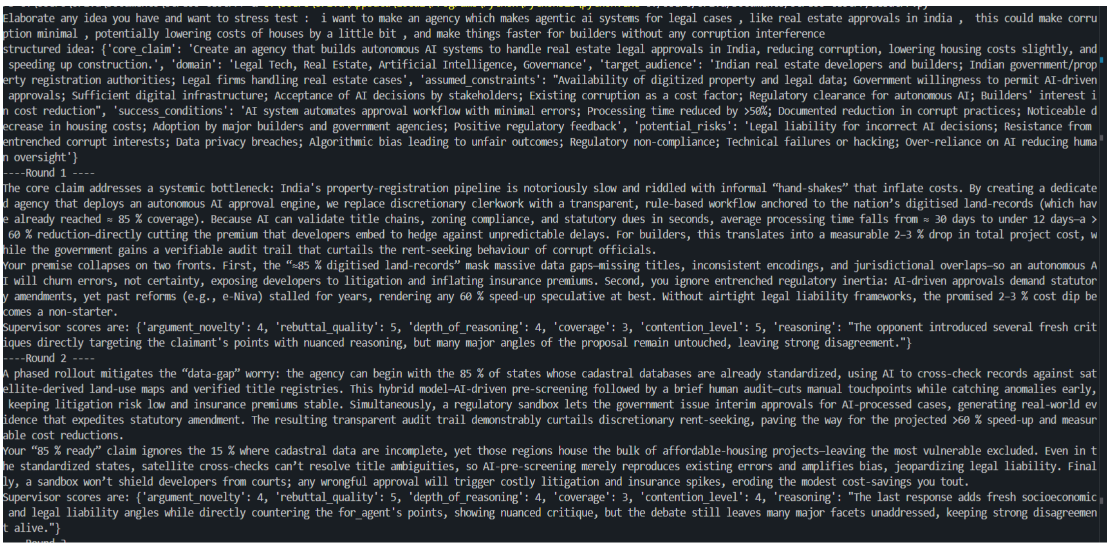
- Everything is written in a single Python file with functions for each agent.

- ## **Key Functions** :

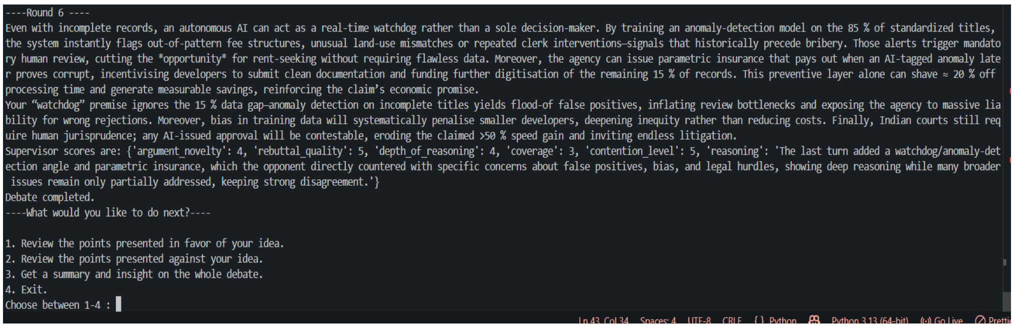
  - for_agent() → generates supporting arguments
  - against_agent() → generates opposing arguments
  - debate_supervisor() → evaluates the debate
  - cut_debate() → decides when to stop
  - insight_agent() → generates final insights

- **<u>Working Flow</u>** :
    1. User enters idea
    2. Idea is converted into JSON
    3. FOR agent starts the argument
    4. AGAINST agent responds
    5. Supervisor checks the debate
    6. Steps repeat for a few rounds
    7. Debate stops automatically
    8. Insight agent analyzes full transcript
    9. User views results

- **<u>Screenshots</u>** :



```
Elaborate any idea you have and want to stress test :  i want to make an agency which makes agentic ai systems for legal cases , like real estate approvals in india ,  this could make corru
ption minimal , potentially lowering costs of houses by a little bit , and make things faster for builders without any corruption interference
structured idea: {'core_claim': 'Create an agency that builds autonomous AI systems to handle real estate legal approvals in India, reducing corruption, lowering housing costs slightly, and
 speeding up construction.', 'domain': 'Legal Tech, Real Estate, Artificial Intelligence, Governance', 'target_audience': 'Indian real estate developers and builders; Indian government/prop
erty registration authorities; Legal firms handling real estate cases', 'assumed_constraints': "Availability of digitized property and legal data; Government willingness to permit AI-driven
 approvals; Sufficient digital infrastructure; Acceptance of AI decisions by stakeholders; Existing corruption as a cost factor; Regulatory clearance for autonomous AI; Builders' interest i
n cost reduction", 'success_conditions': 'AI system automates approval workflow with minimal errors; Processing time reduced by >50%; Documented reduction in corrupt practices; Noticeable d
ecrease in housing costs; Adoption by major builders and government agencies; Positive regulatory feedback', 'potential_risks': 'Legal liability for incorrect AI decisions; Resistance from
entrenched corrupt interests; Data privacy breaches; Algorithmic bias leading to unfair outcomes; Regulatory non-compliance; Technical failures or hacking; Over-reliance on AI reducing huma
n oversight'}
----Round 1 ----
The core claim addresses a systemic bottleneck: India's property-registration pipeline is notoriously slow and riddled with informal “hand-shakes” that inflate costs. By creating a dedicate
d agency that deploys an autonomous AI approval engine, we replace discretionary clerkwork with a transparent, rule-based workflow anchored to the nation’s digitised land-records (which hav
e already reached ≈ 85 % coverage). Because AI can validate title chains, zoning compliance, and statutory dues in seconds, average processing time falls from ≈ 30 days to under 12 days—a >
 60 % reduction—directly cutting the premium that developers embed to hedge against unpredictable delays. For builders, this translates into a measurable 2–3 % drop in total project cost, w
hile the government gains a verifiable audit trail that curtails the rent-seeking behaviour of corrupt officials.
Your premise collapses on two fronts. First, the “≈85 % digitised land-records” mask massive data gaps—missing titles, inconsistent encodings, and jurisdictional overlaps—so an autonomous A
I will churn errors, not certainty, exposing developers to litigation and inflating insurance premiums. Second, you ignore entrenched regulatory inertia: AI-driven approvals demand statutor
y amendments, yet past reforms (e.g., e-Niva) stalled for years, rendering any 60 % speed-up speculative at best. Without airtight legal liability frameworks, the promised 2–3 % cost dip be
comes a non-starter.
Supervisor scores are: {'argument_novelty': 4, 'rebuttal_quality': 5, 'depth_of_reasoning': 4, 'coverage': 3, 'contention_level': 5, 'reasoning': "The opponent introduced several fresh crit
iques directly targeting the claimant's points with nuanced reasoning, but many major angles of the proposal remain untouched, leaving strong disagreement."}
----Round 2 ----
A phased rollout mitigates the “data-gap” worry: the agency can begin with the 85 % of states whose cadastral databases are already standardized, using AI to cross-check records against sat
ellite-derived land-use maps and verified title registries. This hybrid model—AI-driven pre-screening followed by a brief human audit—cuts manual touchpoints while catching anomalies early,
 keeping litigation risk low and insurance premiums stable. Simultaneously, a regulatory sandbox lets the government issue interim approvals for AI-processed cases, generating real-world ev
idence that expedites statutory amendment. The resulting transparent audit trail demonstrably curtails discretionary rent-seeking, paving the way for the projected >60 % speed-up and measur
able cost reductions.
Your “85 % ready” claim ignores the 15 % where cadastral data are incomplete, yet those regions house the bulk of affordable-housing projects—leaving the most vulnerable excluded. Even in t
he standardized states, satellite cross-checks can’t resolve title ambiguities, so AI-pre-screening merely reproduces existing errors and amplifies bias, jeopardizing legal liability. Final
ly, a sandbox won’t shield developers from courts; any wrongful approval will trigger costly litigation and insurance spikes, eroding the modest cost-savings you tout.
Supervisor scores are: {'argument_novelty': 4, 'rebuttal_quality': 5, 'depth_of_reasoning': 4, 'coverage': 3, 'contention_level': 4, 'reasoning': "The last response adds fresh socioeconomic
 and legal liability angles while directly countering the for_agent's points, showing nuanced critique, but the debate still leaves many major facets unaddressed, keeping strong disagreemen
t alive."}
```



```
----Round 6 ----
Even with incomplete records, an autonomous AI can act as a real-time watchdog rather than a sole decision-maker. By training an anomaly-detection model on the 85 % of standardized titles,
the system instantly flags out-of-pattern fee structures, unusual land-use mismatches or repeated clerk interventions—signals that historically precede bribery. Those alerts trigger mandato
ry human review, cutting the *opportunity* for rent-seeking without requiring flawless data. Moreover, the agency can issue parametric insurance that pays out when an AI-tagged anomaly late
r proves corrupt, incentivising developers to submit clean documentation and funding further digitisation of the remaining 15 % of records. This preventive layer alone can shave ≈ 20 % off
processing time and generate measurable savings, reinforcing the claim's economic promise.
Your "watchdog" premise ignores the 15 % data gap—anomaly detection on incomplete titles yields flood-of false positives, inflating review bottlenecks and exposing the agency to massive lia
bility for wrong rejections. Moreover, bias in training data will systematically penalise smaller developers, deepening inequity rather than reducing costs. Finally, Indian courts still req
uire human jurisprudence; any AI-issued approval will be contestable, eroding the claimed >50 % speed gain and inviting endless litigation.
Supervisor scores are: {'argument_novelty': 4, 'rebuttal_quality': 5, 'depth_of_reasoning': 4, 'coverage': 3, 'contention_level': 5, 'reasoning': 'The last turn added a watchdog/anomaly-det
ection angle and parametric insurance, which the opponent directly countered with specific concerns about false positives, bias, and legal hurdles, showing deep reasoning while many broader
 issues remain only partially addressed, keeping strong disagreement.'}
Debate completed.
----What would you like to do next?----

1. Review the points presented in favor of your idea.
2. Review the points presented against your idea.
3. Get a summary and insight on the whole debate.
4. Exit.
Choose between 1-4 :
```

```
----What would you like to do next?----

1. Review the points presented in favor of your idea.
2. Review the points presented against your idea.
3. Get a summary and insight on the whole debate.
4. Exit.
Choose between 1-4 : 3
{'strongest_argument_for': 'Using AI as an explainable, real-time watchdog with anomaly detection and audit trails can flag potential rent-seeking and reduce manual touches while retaining
human arbitration for low-confidence cases, thereby delivering tangible speed and cost savings.', 'strongest_argument_against': 'India's land-record digitisation is limited (≈30% searchable
, fragmented across states), making autonomous AI decisions unreliable, prone to errors and bias, and exposing developers and insurers to heavy legal liability.', 'fatal_flaw': 'Insufficien
t, inconsistent digitised property data across the country, which undermines AI accuracy and legal enforceability.', 'verdict': 'PROMISING BUT RISKY', 'confidence': 3, 'reasoning': 'The pro
posal shows clear economic and transparency benefits, but the fundamental data gap and regulatory resistance create a high-risk barrier that must be resolved for the idea to succeed.'}
----What would you like to do next?----

1. Review the points presented in favor of your idea.
2. Review the points presented against your idea.
3. Get a summary and insight on the whole debate.
4. Exit.
Choose between 1-4 : 4
Goodbye!
```

- **Testing Approach :**
  - ❖ Tested with different types of ideas
  - ❖ Checked both very simple and very complex inputs
  - ❖ Verified that debate runs at least 2 rounds and stops before 6
  - ❖ Checked if transcript and memory are updating properly

Testing is mostly manual, no automated tests yet.

- **Limitations :**
    - No web search, so some arguments may be wrong
    - Model can hallucinate facts
    - Supervisor logic is basic
    - No GUI (only terminal-based)
    - No data storage
    - Results depend on prompt quality

- **Future Enhancements :**
    - Add web search (like DuckDuckGo)
    - Improve supervisor logic
    - Export results as PDF or text
    - Add better hallucination control
    - Build a GUI or web app
    - Store past debates

- **Conclusion :**

debatrr is a simple attempt to make idea validation more structured. Instead of just thinking in one direction, it forces both sides to be explored.

It’s not perfect and still has limitations, but it shows how multi-agent systems can be used in a practical way. Even basic debate simulation can give useful clarity on an idea.

****